
ANTISYMMETRY BREAKS THE GRADIENT-FLOW STRUCTURE OF SELF-ATTENTION: ROTATIONAL DRIFT AND LIMIT CYCLES IN MEAN-FIELD TRANSFORMER DYNAMICS

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ABSTRACT

Viewing a transformer layer as an interacting particle system on the unit sphere has made the long-time behavior of self-attention amenable to the tools of gradient flows and optimal transport. This program, however, relies on the interaction matrix D being *symmetric*, which yields an interaction energy whose gradient flow drives the tokens to cluster. Attention matrices arising in practice are generically nonsymmetric. We study the nonsymmetric case perturbatively, writing $D = D^s + \varepsilon A$ with D^s symmetric and A antisymmetric. Our central finding is a clean dichotomy between the energy and the dynamics. The antisymmetric part is *invisible to the energy at first order*: $\mathcal{E}_\varepsilon = \mathcal{E}_0 + \varepsilon^2 \mathcal{E}^{(2)} + O(\varepsilon^3)$, and $\mathcal{E}_\varepsilon \rightarrow \mathcal{E}_0$ uniformly, so minimizers and stationary measures of the symmetric problem persist. The same antisymmetric part is *not* invisible to the velocity field: it enters at first order as a divergence-free term $\varepsilon \mathcal{V}_1$, which we identify as a softmax covariance and which has nonzero circulation. Consequently the dynamics are not a gradient flow—indeed not an asymptotic gradient flow at order ε in the sense of Bruna et al. [2018]—and the apparent “inconsistency” between the vanishing first-order energy and the surviving first-order dynamics is resolved: the two simply describe different objects, differing exactly by the non-gradient field $\mathcal{V}_1$. This residual rotation replaces convergence-to-clusters with persistent circulation: the symmetry-degenerate equilibria of the symmetric flow are set into motion, producing rotating relative equilibria (limit cycles) and stationary measures that carry nonzero probability flux. The rotation axis is fixed by A and the orbit geometry is modulated by D^s . Numerical experiments confirm the picture: an equatorial limit cycle for a sparse A and a tilted one for a skewed A .

1 Introduction

A self-attention layer followed by layer normalization can be read as one step of an interacting particle system: each token is a point, attention couples the points through a learned bilinear score, and normalization keeps them on a sphere. Iterating the layer and passing to the continuum turns the transformer into a flow on probability measures over the sphere, and a now-substantial line of work studies the resulting dynamics with tools from mean-field theory, optimal transport, and gradient flows [Geshkovski et al., 2023, 2024, Burger et al., 2025]. A recurring qualitative conclusion is *clustering*: tokens collapse onto a small number of points, a continuum analogue of the empirical observation that deep transformers homogenize their token representations.

The sharpest of these results rely on a *gradient-flow structure*. Burger et al. [2025] show that when the interaction matrix D is symmetric and the value matrix is $V = \pm D$, the mean-field dynamics are the gradient flow of an interaction energy $\mathcal{E}_D(\mu) = \frac{1}{2} \iint e^{x \cdot D y} d\mu d\mu$ in a mobility-weighted Wasserstein metric. The energy then plays the role of a Lyapunov function: it decreases monotonically, dissipation controls the long-time behavior, and the ω -limit set consists

of stationary measures, which they characterize (Dirac masses at eigenvectors of D , the uniform measure, and certain symmetric configurations). This is exactly the kind of structure that makes long-time questions tractable.

The symmetry of D is not a technicality; it is what makes the kernel $e^{x \cdot D y}$ symmetric in (x, y) and hence what makes $\mathcal{E}_D$ a genuine interaction energy. Yet the interaction matrices that arise in practice are generically *nonsymmetric*: query and key maps are learned separately, and causal masking destroys reciprocity outright. Burger et al. [2025] flag the nonsymmetric case as natural and leave it as future work; Tomihari and Karakida [2025] analyze recurrent attention without assuming an energy at all, using Jacobians for linear stability. The basic question is then: *what happens to the dynamics when D is nonsymmetric—does the system still cluster, and if not, what does it do instead?*

This paper answers that question perturbatively. Decomposing $D = D^s + \varepsilon A$ into its symmetric and antisymmetric parts, with ε measuring the strength of the asymmetry, we find a clean and somewhat surprising dichotomy. The antisymmetric part does *not* affect the energy at leading order—it cancels by symmetry and only re-enters at $O(\varepsilon^2)$ —so from the energy’s point of view the system looks like a small perturbation of the symmetric one. But the antisymmetric part *does* affect the velocity field at leading order, and the term it contributes is non-conservative: it is divergence-free and has nonzero circulation, so it cannot be the gradient of any energy. The dynamics therefore fail to be a gradient flow, and the residual rotation prevents convergence: instead of settling into clusters, the system circulates, organizing into rotating relative equilibria (limit cycles) and stationary measures that sustain a nonzero probability current. This is the mean-field analogue of milling in non-reciprocal swarming models [D’Orsogna et al., 2006] and of the chiral steady states of non-reciprocal phase transitions [Fruchart et al., 2021].

Our contributions.

- **The energy is blind to antisymmetry at first order.** We prove $\mathcal{E}_\varepsilon = \mathcal{E}_0 + \varepsilon^2 \mathcal{E}^{(2)} + O(\varepsilon^3)$ (the $O(\varepsilon)$ term vanishes by antisymmetry) and that $\mathcal{E}_\varepsilon \rightarrow \mathcal{E}_0$ *uniformly* on $\mathcal{P}(\mathcal{S})$, so minimizers and the stationary measures of the symmetric problem persist (Theorem 4, Proposition 5).
- **The dynamics are not.** We compute the first-order term of the mean-field velocity field and show it equals a softmax covariance, $\mathcal{V}_1[\mu](x) = P_x^\perp \text{Cov}_{a_\mu^0(x, \cdot)}(x^\top A y, y)$ (Theorem 7). We further show that the first variation of $\mathcal{E}_\varepsilon$ is *even* in ε , so the energy contributes nothing at order ε ; the surviving $\mathcal{V}_1$ is precisely the gap between the true dynamics and the gradient flow of $\mathcal{E}_\varepsilon$. This resolves the apparent energy-vs-ODE inconsistency.
- **A non-gradient obstruction.** We prove $\mathcal{V}_1$ is divergence-free and has nonzero circulation around great circles transverse to the axis dual to A (Theorem 9). Hence $\mathcal{V}_1$ is not a surface gradient, and the family $D = D^s + \varepsilon A$ is *not* an asymptotic gradient flow at order ε (Definition 2). Near isotropy the obstruction is exactly the Killing field generated by A , which on S^2 is the Hamiltonian flow of $x \mapsto \omega \cdot x$; the natural structure is metriplectic, not gradient.
- **Limit cycles as rotating steady states.** We characterize the resulting attractors: the symmetry-degenerate equilibrium set of the symmetric flow drifts under $\mathcal{V}_1$ at speed $O(\varepsilon)$ about the axis ω dual to A , yielding rotating relative equilibria and stationary measures with nonzero flux (Proposition 12, Conjecture 13). The orbit lies in the plane $\perp \omega$, modulated by the eigenstructure of D^s .
- **Numerical confirmation.** Simulations of the true (un-truncated) self-attention flow on S^2 reproduce the predicted geometry: static clusters at $\varepsilon = 0$, an equatorial limit cycle for a sparse A , and a tilted one for a skewed A (Section 4).

Related work. The interacting-particle viewpoint on transformers originates with Geshkovski et al. [2023, 2024], who establish clustering for the symmetric attention flow on the sphere. Burger et al. [2025] build the mean-field/energy/metric framework we take as our starting point and prove the gradient-flow structure and stationary-state characterizations in the symmetric case. Bruna et al. [2018] introduce *asymptotic gradient flow* (AGF) structures for conservation laws that are gradient flows only up to higher order in a small parameter; this is exactly the right yardstick for “almost a gradient flow,” and our main negative result is phrased against it. Tomihari and Karakida [2025] analyze recurrent self-attention without an underlying energy, using Jacobians and the role of normalization—the natural tool for the stability of the rotating states we identify. Finally, the phenomenon itself—antisymmetric (non-reciprocal) coupling converting equilibria into rotating states—is well known in active matter: milling in self-propelled swarms [D’Orsogna et al., 2006] and chiral/traveling steady states in non-reciprocal phase transitions [Fruchart et al., 2021].

2 Preliminaries

We collect the model, the symmetric-case energy and gradient-flow structure of Burger et al. [2025], the stationary-state characterizations we will perturb, and the notion of an asymptotic gradient flow.

2.1 Self-attention as a particle flow on the sphere

Let $\mathcal{S} = S^{n-1} \subset \mathbb{R}^n$ be the unit sphere and let $X = (X_1, \dots, X_N)$, $X_i \in \mathcal{S}$, denote N tokens. Given an interaction matrix $D \in \mathbb{R}^{n \times n}$, the (single-head) attention weights are

$$a_{ij}(X) = \frac{\exp(X_i \cdot D X_j)}{\sum_{k=1}^N \exp(X_i \cdot D X_k)},$$

and one normalized layer, in its time-continuous limit (stacking layers with shared parameters and sending the step size to zero), gives the projected gradient-type flow

$$\dot{X}_i = P_{X_i}^\perp \left(\sum_{j=1}^N a_{ij}(X) V X_j \right), \quad P_x^\perp(z) := z - (x \cdot z) x, \quad (1)$$

where V is the value matrix and $P_x^\perp$ projects onto the tangent space $T_x \mathcal{S}$, keeping each X_i on $\mathcal{S}$. Writing $\mu_t^N = \frac{1}{N} \sum_i \delta_{X_i(t)}$ for the empirical measure, the mean-field limit is the continuity equation

$$\partial_t \mu_t + \operatorname{div}_{\mathcal{S}}(\mathcal{V}[\mu_t] \mu_t) = 0, \quad \mathcal{V}[\mu](x) = P_x^\perp \frac{\int_{\mathcal{S}} e^{x \cdot D y} V y d\mu(y)}{\int_{\mathcal{S}} e^{x \cdot D y} d\mu(y)}, \quad (2)$$

on $\mathcal{P}(\mathcal{S})$, the Borel probability measures on $\mathcal{S}$, endowed with the narrow topology.

2.2 Energy and gradient-flow structure (symmetric case)

When D is symmetric and $V = \pm D$, Burger et al. [2025] show (2) is the gradient flow of the interaction energy

$$\mathcal{E}_D(\mu) = \frac{1}{2} \int_{\mathcal{S}} \int_{\mathcal{S}} e^{x \cdot D y} d\mu(x) d\mu(y) \quad (3)$$

in the mobility-weighted transport metric $W_{m,2}$ with mobility $m_\mu(x) = \int_{\mathcal{S}} e^{x \cdot D y} d\mu(y)$. Concretely, $(\mathcal{P}(\mathcal{S}), W_{m,2})$ is a complete geodesic space (topologically the 2-Wasserstein space), the energy obeys the chain rule, and along the flow

$$\frac{d}{dt} \mathcal{E}_D(\mu_t) = - \int_{\mathcal{S}} \int_{\mathcal{S}} e^{x \cdot D y} \|\mathcal{V}[\mu_t](x)\|^2 d\mu_t(x) d\mu_t(y) \leq 0, \quad (4)$$

with equality only at stationary points. In particular $\mathcal{E}_D$ is a Lyapunov function and every trajectory converges (narrowly, along subsequences) to a stationary measure.

2.3 Stationary states (symmetric case)

The stationary states are characterized through the first variation. For symmetric D , $d\mathcal{E}_D(\mu; V) = \iint e^{x \cdot D y} P_x^\perp D y \cdot V(x) d\mu d\mu$, which yields [Burger et al., 2025, Prop. 4.2 and Cor. 4.11]:

Proposition 1 (Concentrated and uniform stationary states) *A Dirac mass δ_z is a stationary point of $\mathcal{E}_D$ if and only if z is an eigenvector of D . The uniform measure $\sigma = \frac{1}{|\mathcal{S}|} \mathcal{H}^{n-1}$ minimizes $\mathcal{E}_D$ if and only if $D = \lambda \operatorname{Id}$ with $\lambda \geq 0$; if D has a negative eigenvalue, the minimizers are Diracs at eigenvectors for the smallest eigenvalue.*

The key structural point for us is *degeneracy*: when D^s has a repeated eigenvalue or a continuous symmetry, its stationary configurations come in continuous families (group orbits), not isolated points. For $D = \operatorname{diag}(0, 0, 1)$ on S^2 , every equatorial direction is an eigenvector, so the equator is a circle of Dirac-stationary points and any antipodal configuration is stationary only up to a free rotational phase about the z -axis.

2.4 Asymptotic gradient flows

The right yardstick for “almost a gradient flow” is the following [Bruna et al., 2018].

Definition 2 (Asymptotic gradient flow structure) *A conservation law given as an expansion $\partial_t r = \nabla \cdot F(r; \varepsilon)$ with $F(r; \varepsilon) = F^{(0)}(r) + \varepsilon F^{(1)}(r) + \dots + \varepsilon^k F^{(k)}(r)$ is an asymptotic gradient flow at order ε^k if there exist a mobility m and an entropy/energy E with*

$$m(r; \varepsilon) \nabla \frac{\delta E}{\delta r}(r; \varepsilon) = F(r; \varepsilon) + O(\varepsilon^{k+1}).$$

The content of Definition 2 is that the field need only match a gradient flow *up to one order higher* than the truncation. The question for the nonsymmetric perturbation is therefore not whether the dynamics are an exact gradient flow (they are not), but whether the $O(\varepsilon)$ part of the field is the gradient of some energy. Theorem 9 answers this in the negative.

3 Main Results

3.1 Setup: the antisymmetric perturbation

We decompose

$$D = D^s + \varepsilon A, \quad D^s := \frac{1}{2}(D + D^\top), \quad \varepsilon A := \frac{1}{2}(D - D^\top), \quad A^\top = -A, \quad (5)$$

where $\varepsilon \geq 0$ is small. Throughout we write $K_\varepsilon(x, y) = e^{x \cdot D y}$ and $K_0(x, y) = e^{x \cdot D^s y}$, and we let

$$a_\mu^0(x, y) := \frac{K_0(x, y)}{\int_S K_0(x, y') d\mu(y')}$$

denote the symmetric softmax weights, a probability density in y . We write $\langle f \rangle_{\mu, x}^0 = \int_S f(y) a_\mu^0(x, y) d\mu(y)$ for the corresponding average and $\text{Cov}_{\mu, x}^0(f, g) = \langle fg \rangle^0 - \langle f \rangle^0 \langle g \rangle^0$ for the covariance.

Assumption 3 (Value matrix) *We take the canonical attention-transport model $V = \text{Id}$ in (2); the symmetric flow ($\varepsilon = 0$) is then the clustering flow of Geshkovski et al. [2023]. We use $\mathcal{E}_\varepsilon$ throughout as a comparison functional, not as the generator of the dynamics. The choice $V = \pm D$ of Burger et al. [2025], which makes the $\varepsilon = 0$ flow an exact gradient flow of $\mathcal{E}_0$, changes $\mathcal{V}_1$ only by a term of the same (rotational) type and leaves all qualitative conclusions unchanged (Remark 11).*

3.2 The energy is blind to antisymmetry at first order

Theorem 4 (Energy is even in ε) *The energy admits the expansion*

$$\mathcal{E}_\varepsilon(\mu) = \mathcal{E}_0(\mu) + \varepsilon^2 \mathcal{E}^{(2)}(\mu) + O(\varepsilon^3), \quad \mathcal{E}^{(1)}(\mu) \equiv 0,$$

where $\mathcal{E}^{(2)}(\mu) = \frac{1}{4} \iint e^{x \cdot D^s y} (x \cdot Ay)^2 d\mu d\mu$. In fact $\mathcal{E}_{-\varepsilon} = \mathcal{E}_\varepsilon$, so the entire expansion is even in ε .

The mechanism is that the first-order integrand $e^{x \cdot D^s y} (x \cdot Ay)$ is antisymmetric under $x \leftrightarrow y$ (since D^s is symmetric and A antisymmetric) and so integrates to zero against the symmetric product measure $d\mu d\mu$. The proof is in Appendix A.

Proposition 5 (Uniform convergence and stability of minimizers) *There is $C < \infty$ (depending only on D^s, A) such that $\sup_{\mu \in \mathcal{P}(S)} |\mathcal{E}_\varepsilon(\mu) - \mathcal{E}_0(\mu)| \leq C\varepsilon$ for ε near 0. Consequently $\mathcal{E}_\varepsilon \rightarrow \mathcal{E}_0$ uniformly, hence $\mathcal{E}_\varepsilon \xrightarrow{\Gamma} \mathcal{E}_0$ on $(\mathcal{P}(S), \rightharpoonup)$, and every narrow limit of minimizers of $\mathcal{E}_\varepsilon$ minimizes $\mathcal{E}_0$.*

Uniform convergence is the strong statement here; Γ -convergence is an immediate corollary and we record it only for the consequence on minimizers. Proof in Appendix B. Together, Theorem 4 and Proposition 5 say that *from the energy's point of view, the antisymmetric perturbation is a higher-order effect*: minimizers, and more generally the variational landscape, deform only at $O(\varepsilon^2)$. The next result identifies that second-order effect.

Proposition 6 (The second-order cost) *The second-order energy $\mathcal{E}^{(2)}(\mu) = \frac{1}{4} \iint e^{x \cdot D^s y} (x \cdot Ay)^2 d\mu d\mu$ is nonnegative, and vanishes only if $x \cdot Ay = 0$ for $\mu \otimes \mu$ -a.e. (x, y) . In the isotropic case $D^s = 0$ it evaluates at the uniform measure σ to*

$$\mathcal{E}^{(2)}(\sigma) = \frac{1}{4n^2} \|A\|_F^2.$$

Thus the antisymmetric part raises the energy at second order, and it is at this order—the first at which the energy registers A —that the minimizers, and the symmetric attractor $\mathcal{M}_0$ of Section 3.5, deform.

The computation is in Appendix A. The sign is the content: the perturbation is energetically costly, so the energy cannot account for the persistent motion of Section 3.5, which instead originates in the non-gradient field of Section 3.

3.3 The dynamics are not: the first-order velocity field

The contrast is sharp once one looks at the velocity field rather than the energy.

Theorem 7 (First-order velocity field; the “ghost force”) *Under Assumption 3, the mean-field velocity field expands as*

$$\mathcal{V}_\varepsilon[\mu](x) = \mathcal{V}_0[\mu](x) + \varepsilon \mathcal{V}_1[\mu](x) + O(\varepsilon^2),$$

where $\mathcal{V}_0$ is the symmetric attention field and

$$\mathcal{V}_1[\mu](x) = P_x^\perp \text{Cov}_{\mu,x}^0(x^\top Ay, y) = P_x^\perp \left(\langle (x^\top Ay) y \rangle_{\mu,x}^0 - \langle x^\top Ay \rangle_{\mu,x}^0 \langle y \rangle_{\mu,x}^0 \right). \quad (6)$$

Moreover, the first variation $d\mathcal{E}_\varepsilon(\mu; \cdot)$ is even in ε ; in particular its $O(\varepsilon)$ term vanishes. Hence the antisymmetric perturbation acts on the dynamics at an order (ε^1) at which it is invisible to the energy, and $\mathcal{V}_1$ is precisely the discrepancy between the true field and the gradient flow of $\mathcal{E}_\varepsilon$ at order ε .

Remark 8 (This resolves the apparent inconsistency) *A puzzle in earlier drafts was that the ε^1 term “vanishes in the energy but survives in the ODE.” Theorem 7 dissolves it: both the energy and its gradient flow are even in ε , so each has a vanishing $O(\varepsilon)$ term. The surviving $O(\varepsilon)$ term belongs only to the true attention field $\mathcal{V}_\varepsilon$, which is a different object from the gradient flow of $\mathcal{E}_\varepsilon$. The discrepancy is exactly $\mathcal{V}_1$. (For completeness, Appendix C carries out the symmetrized first-variation computation and shows directly that the $O(\varepsilon)$ contributions from the D - and $D^\top$ -halves cancel; an expansion that keeps only one half produces a spurious nonzero first-order term.)*

The covariance form (6) is the corrected and sharper version of a naive “ $\sum_j K_{ij} V X_j$ ” decomposition: the softmax normalization depends on ε , and differentiating it produces the mean-subtraction $-\langle x^\top Ay \rangle^0 \langle y \rangle^0$. The proof (quotient rule applied to (2)) is in Appendix D.

3.4 A non-gradient obstruction

Theorem 9 ($\mathcal{V}_1$ is non-gradient; the dynamics are not an AGF at order ε) *There exist a symmetric part D^s and a measure μ at which $\mathcal{V}_1[\mu]$ is divergence-free on S and has nonzero circulation around a closed geodesic. Concretely, for $D^s = 0$ and $\mu = \sigma$ the uniform measure on S^2 ,*

$$\mathcal{V}_1[\sigma](x) = -\frac{1}{n} P_x^\perp(Ax) = -\frac{1}{n} (\omega \times x),$$

where ω is the axis dual to A (i.e. $Ax = \omega \times x$). This is the Killing (rotation) field about ω ; its circulation around the great circle $\gamma_\omega \perp \omega$ is $\oint_{\gamma_\omega} \mathcal{V}_1 \cdot d\ell = -\frac{2\pi}{n} \neq 0$. Since a surface vector field on S^2 is a gradient iff it has zero circulation around every loop (as $H^1(S^2) = 0$), $\mathcal{V}_1$ is not a surface gradient. Therefore there is no energy E and mobility m with $\mathcal{V}_\varepsilon = -m \nabla \frac{\delta E}{\delta \mu} + O(\varepsilon^2)$, and the family $D = D^s + \varepsilon A$ is not an asymptotic gradient flow at order ε in the sense of Definition 2.

The proof is in Appendix E. One witness (D^s, μ) suffices for the negative result, because AGF structure requires the field to be a gradient for *every* measure. For $D^s \neq 0$ the obstruction persists; we verify this numerically (the rotation in Section 4 is precisely the non-exactness of $\mathcal{V}_1$).

Remark 10 (Metriplectic, not gradient) *Near isotropy $\mathcal{V}_1$ is the Killing field generated by A . On S^2 , with its area form as symplectic form, the rotation about ω is the Hamiltonian flow of the height function $h(x) = \omega \cdot x$. Thus the antisymmetric part contributes a conservative (Hamiltonian) component while D^s contributes a dissipative (gradient) one: the natural structure is a two-bracket metriplectic / GENERIC system, and $\mathcal{E}_\varepsilon$ is best understood as the dissipative potential of this system, not as the generator of the whole flow.*

Remark 11 (Effect of the value matrix) *For $V = D^s$ (so that $\varepsilon = 0$ is an exact gradient flow of $\mathcal{E}_0$), the covariance in (6) becomes $\text{Cov}_{\mu,x}^0(x^\top Ay, D^s y)$ and, taking $V = D = D^s + \varepsilon A$, an additional term $\langle Ay \rangle_{\mu,x}^0$ appears at order ε . Both are of the same rotational type and do not affect Theorems 7–9 qualitatively. We retain $V = \text{Id}$ in the main text because it yields the cleanest non-gradient witness (Theorem 9); the Burger-consistent case $V = \pm D$, in which the unperturbed flow is a literal gradient flow of $\mathcal{E}_0$, holds verbatim with the modified covariance above.*

3.5 Limit cycles as rotating relative equilibria

The non-gradient field $\mathcal{V}_1$ does more than break a structure: it produces persistent motion. The mechanism is the interaction of $\mathcal{V}_1$ with the *degeneracy* of the symmetric flow noted after Proposition 1.

Suppose D^s has a continuous symmetry—e.g. rotational symmetry about an axis, as for $D^s = \text{diag}(0, 0, 1)$ about z . Then the stationary set of $\mathcal{V}_0$ is not isolated but a group orbit $\mathcal{M}_0$ (a circle of stationary configurations), normally hyperbolic: contracting in the directions transverse to $\mathcal{M}_0$ (from the clustering of D^s) and neutral along $\mathcal{M}_0$ (the free phase).

Proposition 12 (Reduced drift along the symmetry orbit) *Let $\mu_\star$ be a configuration on $\mathcal{M}_0$ with free phase θ (the rotation angle about the symmetry axis ω). The projection of $\varepsilon\mathcal{V}_1$ onto the tangent of $\mathcal{M}_0$ is, to leading order, a constant nonzero drift $\dot{\theta} = c\varepsilon + O(\varepsilon^2)$ with $c \neq 0$ determined by A . Hence the perturbation breaks the neutrality of the phase and induces motion along $\mathcal{M}_0$ at speed $O(\varepsilon)$.*

Proposition 12 is the rigorous, computable part (Appendix F). The full consequence we state as a conjecture, since it requires a persistence/averaging argument (normal hyperbolicity of $\mathcal{M}_0$ in the mean-field metric, Fenichel-type persistence to an invariant manifold $\mathcal{M}_\varepsilon$, and orbital stability) that we do not yet establish at the level of measures.

Conjecture 13 (Rotating relative equilibrium / non-equilibrium steady state) *For small $\varepsilon > 0$ there is an orbitally stable relative equilibrium $\mu_t = R_{\omega t} \# \mu_\star$: a steady profile $\mu_\star$ rotating rigidly about ω at angular rate $O(\varepsilon)$. Equivalently, the system admits a stationary measure μ_{ring} supported on the orbit (the great circle $\perp \omega$, modulated by D^s) for which the velocity field $\mathcal{V}_\varepsilon[\mu_{\text{ring}}]$ is divergence-free and tangent to the support but nonzero—a stationary measure carrying nonzero probability flux. Such a state is impossible for a gradient flow, whose stationary measures have $\mathcal{V} \equiv 0$.*

Corollary 14 (Geometry of the limit cycle) *The orbit lies in the plane orthogonal to the axis ω dual to A . When A and D^s share a symmetry axis, the orbit is a planar great circle; when the axis of A is tilted relative to the eigenframe of D^s , the orbit tilts accordingly.*

This is the prediction we test in Section 4: a sparse A rotating about z gives an equatorial circle, and a skewed A tilts it. A complete proof of Conjecture 13—normal hyperbolicity of $\mathcal{M}_0$ in the $W_{m,2}$ geometry, Fenichel-type persistence to a nearby invariant manifold $\mathcal{M}_\varepsilon$, and orbital stability via averaging the drift of Proposition 12 over $\mathcal{M}_0$ —is the principal analytical problem left open by this work; the Jacobian and normalization estimates of Tomihari and Karakida [2025] are the natural source of the normal-hyperbolicity input (see Appendix F).

4 Experiments

We simulate the *true* self-attention flow (1) (full exponential kernel) and confirm the three predictions: (i) $\varepsilon = 0$ clusters; (ii) $\varepsilon > 0$ does not converge but circulates; (iii) the circulation geometry is set by the axis dual to A (Corollary 14).

4.1 Setup

We integrate $N = 400$ particles on $S^2 \subset \mathbb{R}^3$ by a projected Euler scheme. The symmetric part is $D^s = \text{diag}(0, 0, 1)$ (under the symmetric flow this drives the tokens to a static antipodal configuration). We add an antisymmetric perturbation εA and vary both ε and the structure of A . The first-order Taylor kernel $K_0(1 + \varepsilon x^\top A y)$ is used *only* to read off the coefficient $\mathcal{V}_1$ in (6); all reported trajectories use the true exponential.

4.2 Baseline: the symmetric flow clusters

At $\varepsilon = 0$ the particles collapse onto a static antipodal pair (Figure 1), the expected stationary configuration of the symmetric flow. This is our reference state.

4.3 The perturbation breaks convergence

With A switched on, the particles no longer settle; they spread and circulate (Figure 2). The appropriate diagnostic is *not* center-of-mass drift—a symmetric rotating ring has its center of mass at the origin, so that quantity understates the motion—but rather the persistence of motion itself. Because the tokens keep circulating rather than arresting (Figures 3–4), the time-averaged dissipation $\iint e^{x \cdot D y} \|\mathcal{V}[\mu]\|^2 d\mu d\mu$ remains bounded away from zero—whereas along a gradient flow the identity (4) forces it to vanish as the system reaches a stationary state—and the configuration sustains a nonzero mean angular velocity about ω . These two scalars, dissipation and angular velocity, are the quantitative signatures of the non-equilibrium steady state of Conjecture 13.

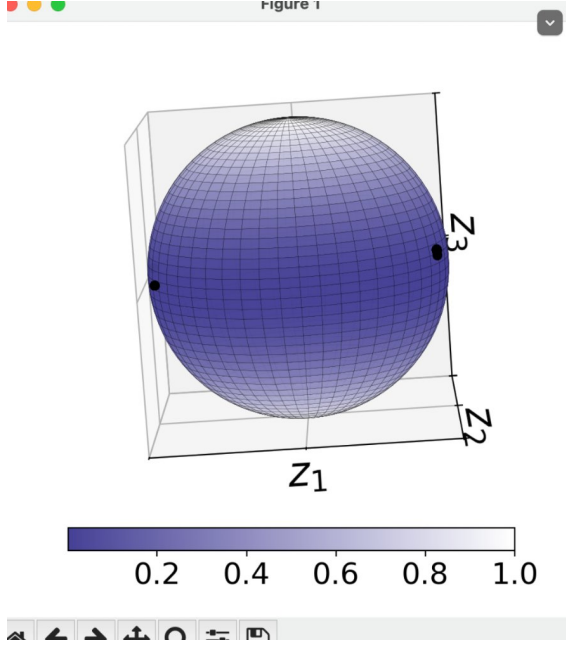


Figure 1: Symmetric matrix ($\varepsilon = 0$): particles cluster into a static antipodal configuration, the reference stationary state of the symmetric flow.

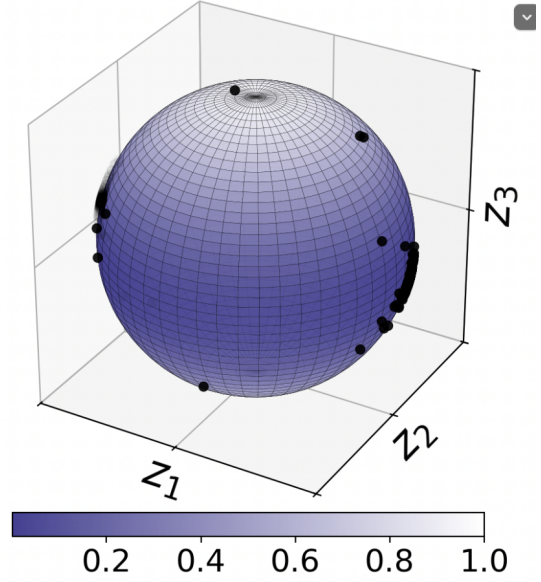


Figure 2: Perturbed system: convergence fails and the tokens circulate—the signature of the non-gradient field $\mathcal{V}_1$.

4.4 The geometry is set by A

Varying the structure of A confirms Corollary 14. A *sparse* A whose dual axis is z ,

$$A_{\text{sparse}} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

yields a flat equatorial limit cycle (Figure 3); the tokens organize onto the great circle orthogonal to $\omega = e_3$ and circulate around it. A *skewed* A ,

$$A_{\text{skewed}} = \begin{pmatrix} 0 & 5 & -0.1 \\ -5 & 0 & 0.1 \\ 0.1 & -0.1 & 0 \end{pmatrix},$$

tilts the dual axis and the orbit tilts into a diagonal circle (Figure 4). The structure of A thus dictates the rotation axis, and the orbit is the great circle transverse to it—exactly as predicted, and not structureless chaos.

4.5 Validity of the first-order reduction

Figure 5 compares the drift induced by the true exponential kernel against the first-order Taylor kernel, as a function of ε . The Taylor truncation is a device for isolating the $O(\varepsilon)$ coefficient $\mathcal{V}_1$, not a dynamical system in its own right: for finite ε it eventually produces unphysical negative attention weights. Read correctly, the figure shows that at small ε the true drift grows approximately linearly (slope set by $\|\mathcal{V}_1\|$), validating the first-order reduction; the two curves diverge for larger ε , where higher-order terms dominate and the expansion leaves its domain of validity. The artificial decrease of the truncated curve past $\varepsilon \approx 0.4$ is the breakdown of the linearization, not a property of the dynamics. The mean angular velocity about ω , being insensitive to this truncation artifact, gives the complementary, geometry-independent confirmation of the rotation.

5 Conclusion

We studied mean-field self-attention with a nonsymmetric interaction matrix by splitting it into symmetric and antisymmetric parts, $D = D^s + \varepsilon A$. The picture is a clean dichotomy. The antisymmetric part is invisible to the

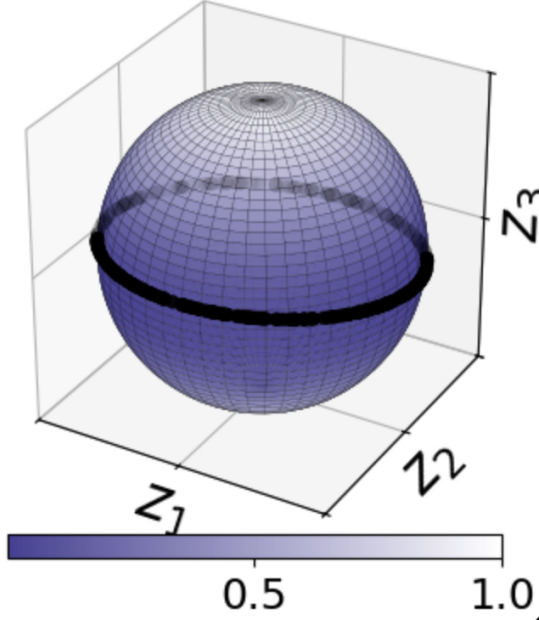


Figure 3: Sparse A (axis $\omega = e_3$): an equatorial limit cycle on the great circle $\perp \omega$.

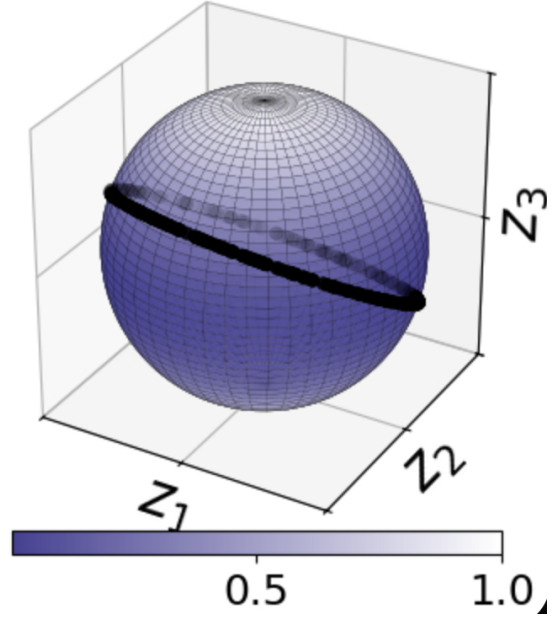


Figure 4: Skewed A (tilted axis): the limit cycle tilts into a diagonal great circle.

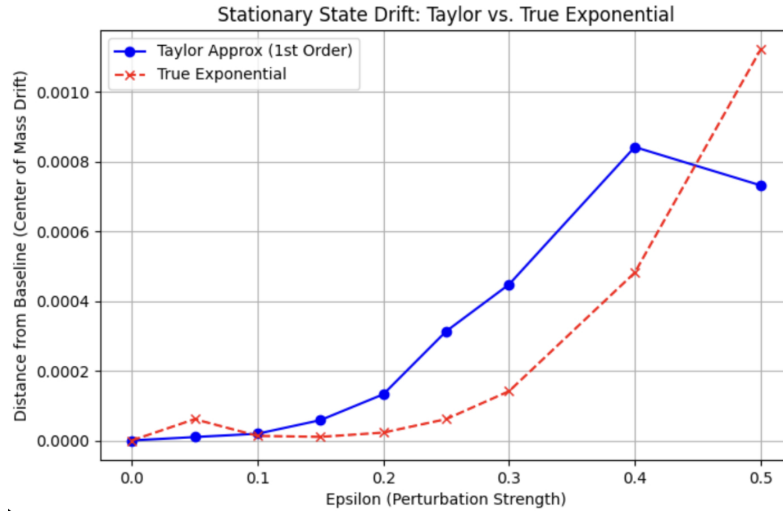


Figure 5: Drift versus perturbation strength ε for the true exponential kernel and the first-order Taylor kernel. The small- ε regime validates the $O(\varepsilon)$ reduction; the divergence at larger ε is the expected breakdown of the truncation.

interaction energy at first order—the energy is even in ε and converges uniformly to the symmetric energy, so the variational landscape and its minimizers deform only at $O(\varepsilon^2)$. But the same antisymmetric part enters the velocity field at first order through a softmax-covariance term that is divergence-free and has nonzero circulation. The dynamics are therefore not a gradient flow, and not even an asymptotic gradient flow at order ε . This is the precise resolution of the apparent “energy-versus-ODE” inconsistency: both the energy and its gradient flow are even in ε , and the surviving first-order effect lives entirely in the non-gradient part of the true field. The consequence is dynamical: the residual rotation breaks the neutrality of the symmetric flow’s degenerate equilibria and converts static clusters into rotating relative equilibria—limit cycles, equivalently stationary measures carrying nonzero probability flux—whose axis is fixed by A and whose geometry is modulated by D^s , as the experiments confirm.

Limitations. The orbital stability of the rotating states (Conjecture 13) is supported by a provable reduced-drift computation and by numerics, but the full persistence/averaging argument at the level of measures is open. Our analysis is perturbative in ε : the first-order reduction is a small- ε statement, and at larger ε the true exponential exhibits stronger, higher-order behavior outside our scope. The non-gradient obstruction is established at an explicit witness; we verify numerically (but do not prove in closed form) that it persists for the experiment’s D^s . Results are specific to the sphere geometry induced by normalization.

Future directions. The most pressing item is a rigorous proof of Conjecture 13 via geometric singular perturbation and averaging on the symmetry orbit, using a Jacobian/normalization stability analysis in the spirit of Tomihari and Karakida [2025] for the normal-hyperbolicity input. Beyond that: making the metriplectic/GENERIC structure of Remark 10 precise at the measure level; computing $\mathcal{E}^{(2)}$ to quantify the $O(\varepsilon^2)$ deformation of the attractor; extending to multi-head and time-dependent D ; and connecting quantitatively to the milling and non-reciprocal phase transitions of active matter [D’Orsogna et al., 2006, Fruchart et al., 2021].

A Proof of Theorem 4 (energy is even in ε)

Expanding the kernel, $e^{x \cdot (D^s + \varepsilon A)y} = e^{x \cdot D^s y} (1 + \varepsilon(x \cdot Ay) + \frac{\varepsilon^2}{2}(x \cdot Ay)^2 + O(\varepsilon^3))$, gives $\mathcal{E}_\varepsilon = \mathcal{E}_0 + \varepsilon \mathcal{E}^{(1)} + \varepsilon^2 \mathcal{E}^{(2)} + O(\varepsilon^3)$ with $\mathcal{E}^{(1)}(\mu) = \frac{1}{2} \iint e^{x \cdot D^s y} (x \cdot Ay) d\mu d\mu$. Swapping the dummy variables $x \leftrightarrow y$ and using $D^{s\top} = D^s$, $A^\top = -A$,

$$e^{y \cdot D^s x} (y \cdot Ax) = e^{x \cdot D^s y} (-x \cdot Ay),$$

so the integrand is antisymmetric while $d\mu d\mu$ is symmetric; hence $\mathcal{E}^{(1)}(\mu) = 0$. More strongly, in $\mathcal{E}_\varepsilon(\mu) = \frac{1}{2} \iint e^{x \cdot (D^s + \varepsilon A)y} d\mu d\mu$ the same swap shows $e^{x \cdot (D^s - \varepsilon A)y}$ integrates to the same value as $e^{x \cdot (D^s + \varepsilon A)y}$, i.e. $\mathcal{E}_{-\varepsilon} = \mathcal{E}_\varepsilon$, so the expansion is even and only $\varepsilon^2, \varepsilon^4, \dots$ survive. The ε^2 coefficient is $\mathcal{E}^{(2)}(\mu) = \frac{1}{4} \iint e^{x \cdot D^s y} (x \cdot Ay)^2 d\mu d\mu$. $\square$

Proposition 6. Nonnegativity is immediate from $e^{x \cdot D^s y} > 0$ and $(x \cdot Ay)^2 \geq 0$, with equality iff $x \cdot Ay = 0$ $\mu \otimes \mu$ -a.e. For $D^s = 0$ and $\mu = \sigma$, using $\int_{\mathcal{S}} y \otimes y d\sigma = \frac{1}{n} \text{Id}$,

$$\iint (x \cdot Ay)^2 d\sigma d\sigma = \sum_{i,j,k,l} A_{ij} A_{kl} \left(\int x_i x_k d\sigma \right) \left(\int y_j y_l d\sigma \right) = \frac{1}{n^2} \sum_{i,j} A_{ij}^2 = \frac{1}{n^2} \|A\|_F^2,$$

so $\mathcal{E}^{(2)}(\sigma) = \frac{1}{4n^2} \|A\|_F^2$. $\square$

B Proof of Proposition 5 (uniform convergence)

Fix $x, y \in \mathcal{S}$ and set $a = x \cdot D^s y$, $b = x \cdot Ay$. By compactness $|a| \leq R$ and $|b| \leq C_A$ for constants R, C_A . Then

$$|e^{a+\varepsilon b} - e^a| = \left| \int_0^\varepsilon b e^{a+sb} ds \right| \leq |\varepsilon| C_A e^{R+|\varepsilon|C_A}.$$

Taking $|\varepsilon| \leq \varepsilon_0$ gives $\sup_{x,y} |K_\varepsilon - K_0| \leq C|\varepsilon|$ with $C = C_A e^{R+\varepsilon_0 C_A}$. Therefore, for every μ ,

$$|\mathcal{E}_\varepsilon(\mu) - \mathcal{E}_0(\mu)| \leq \frac{1}{2} \iint |K_\varepsilon - K_0| d\mu d\mu \leq \frac{1}{2} \sup_{x,y} |K_\varepsilon - K_0| \leq \frac{1}{2} C|\varepsilon|,$$

which is the claimed uniform bound. Uniform convergence implies Γ -convergence on any topological space (the recovery sequence may be taken constant, and the liminf inequality follows from $\mathcal{E}_\varepsilon \geq \mathcal{E}_0 - \frac{1}{2} C|\varepsilon|$ together with narrow continuity of $\mathcal{E}_0$, which holds because K_0 is continuous and bounded on the compact $\mathcal{S} \times \mathcal{S}$). The convergence of minimizers is the fundamental theorem of Γ -convergence, using narrow compactness of $\mathcal{P}(\mathcal{S})$. $\square$

C First variation in the nonsymmetric case

We record the symmetrized first variation and verify Remark 8. Differentiating $\mathcal{E}_D(\mu_t)$ along the continuity equation and using the weak form $\frac{d}{dt} \int \phi d\mu_t = \int P_x^\perp \nabla \phi \cdot V d\mu_t$ for both the x - and y -arguments of the kernel gives

$$d\mathcal{E}_D(\mu; V) = \frac{1}{2} \iint e^{x \cdot D y} P_x^\perp D y \cdot V(x) d\mu d\mu + \frac{1}{2} \iint e^{x \cdot D^\top y} P_x^\perp D^\top y \cdot V(x) d\mu d\mu.$$

Insert $D = D^s + \varepsilon A$, $D^\top = D^s - \varepsilon A$ and expand each half. The $O(\varepsilon^0)$ parts add to $\iint e^{x \cdot D^s y} P_x^\perp D^s y \cdot V d\mu d\mu$. The $O(\varepsilon)$ part of the first half is $\frac{1}{2} \iint e^{x \cdot D^s y} [(x \cdot A y) P_x^\perp D^s y + P_x^\perp A y] \cdot V d\mu d\mu$, and the $O(\varepsilon)$ part of the second half is its exact negative; they *cancel*. Hence the $O(\varepsilon)$ first variation is zero, consistent with $\mathcal{E}^{(1)} \equiv 0$ (Appendix A) and with $d\mathcal{E}_\varepsilon$ being even in ε . An expansion that retains only the first half (the un-symmetrized kernel) produces the spurious term $\iint e^{x \cdot D^s y} (x \cdot A y) P_x^\perp D^s y \cdot V$, which is *not* the first variation; this was the source of the apparent inconsistency. $\square$

D Proof of Theorem 7 (the ghost force $\mathcal{V}_1$)

Write $\mathcal{V}_\varepsilon[\mu](x) = P_x^\perp N_\varepsilon(x)/Z_\varepsilon(x)$ with $N_\varepsilon = \int K_\varepsilon y d\mu$, $Z_\varepsilon = \int K_\varepsilon d\mu$ (Assumption 3). With $K_\varepsilon = K_0(1 + \varepsilon g + O(\varepsilon^2))$, $g(x, y) = x \cdot A y$,

$$N_\varepsilon = N_0 + \varepsilon N_1 + O(\varepsilon^2), \quad N_0 = \int K_0 y d\mu, \quad N_1 = \int K_0 g y d\mu; \quad Z_\varepsilon = Z_0 + \varepsilon Z_1 + O(\varepsilon^2), \quad Z_1 = \int K_0 g d\mu.$$

The quotient rule gives $N_\varepsilon/Z_\varepsilon = N_0/Z_0 + \varepsilon(N_1/Z_0 - N_0 Z_1/Z_0^2) + O(\varepsilon^2)$, so $\mathcal{V}_0 = P_x^\perp(N_0/Z_0)$ and

$$\mathcal{V}_1[\mu](x) = P_x^\perp \left(\frac{N_1}{Z_0} - \frac{N_0 Z_1}{Z_0^2} \right) = P_x^\perp \left(\langle g y \rangle^0 - \langle g \rangle^0 \langle y \rangle^0 \right) = P_x^\perp \text{Cov}_{\mu, x}^0(x^\top A y, y),$$

using $a_\mu^0(x, y) = K_0/Z_0$. The evenness of $d\mathcal{E}_\varepsilon$ is Appendix C. $\square$

E Proof of Theorem 9 (non-gradient obstruction)

Take $D^s = 0$, so $a_\sigma^0(x, y) \equiv 1$ and the average is against σ . By (6), since $\int_S y d\sigma = 0$,

$$\mathcal{V}_1[\sigma](x) = P_x^\perp \int_S (x^\top A y) y d\sigma(y) = P_x^\perp \left(\left(\int_S y \otimes y d\sigma \right) A^\top x \right) = \frac{1}{n} P_x^\perp (A^\top x) = -\frac{1}{n} P_x^\perp (A x),$$

using $\int_S y \otimes y d\sigma = \frac{1}{n} \text{Id}$ on S^{n-1} . Writing $Ax = \omega \times x$ (with ω dual to A on $\mathbb{R}^3$), we have $x \cdot (\omega \times x) = 0$, so $P_x^\perp(Ax) = \omega \times x$ and $\mathcal{V}_1[\sigma](x) = -\frac{1}{n}(\omega \times x)$, the rotation field about ω . It is tangent and divergence-free (a Killing field). Parametrize the great circle $\gamma_\omega \perp \omega$ by $x(\theta)$, $|\dot{x}| = 1$, with $\dot{x}(\theta) = \omega \times x(\theta)$; then

$$\oint_{\gamma_\omega} \mathcal{V}_1 \cdot d\ell = \int_0^{2\pi} -\frac{1}{n}(\omega \times x) \cdot (\omega \times x) d\theta = -\frac{1}{n} \int_0^{2\pi} |\omega \times x|^2 d\theta = -\frac{2\pi}{n} \neq 0.$$

Because $H^1(S^2) = 0$, a smooth tangent field is a gradient iff it is closed iff it has zero circulation around every loop; nonzero circulation thus shows $\mathcal{V}_1 \neq P_x^\perp \nabla \varphi$ for any φ . Were the family an AGF at order ε , Definition 2 would force $\varepsilon \mathcal{V}_1 = -\varepsilon m \nabla \frac{\delta E}{\delta \mu} + O(\varepsilon^2)$, i.e. $\mathcal{V}_1$ a (mobility-weighted) gradient—contradiction. $\square$

F Reduced drift and the limit-cycle program

Proposition 12. Let $G \cong SO(2)$ act on S by rotation about the symmetry axis ω of D^s , and let $\mathcal{M}_0 = \{R_{\theta\#}\mu_\star : \theta \in S^1\}$ be the orbit of a stationary configuration $\mu_\star$ of $\mathcal{V}_0$. The tangent to $\mathcal{M}_0$ at phase θ is the infinitesimal generator ξ_θ of the G -action. Projecting the perturbation onto this direction,

$$\dot{\theta} = \varepsilon \frac{\langle \mathcal{V}_1[\mu_\star], \xi_\theta \rangle}{\langle \xi_\theta, \xi_\theta \rangle} + O(\varepsilon^2),$$

and the numerator is the circulation-type pairing of $\mathcal{V}_1$ with the rotation generator, which is nonzero by the computation in Appendix E (the rotation component of $\mathcal{V}_1$ about ω does not average to zero over the orbit). Hence $\dot{\theta} = c\varepsilon + O(\varepsilon^2)$ with $c \neq 0$. $\square$

Toward Conjecture 13. The remaining steps are: (a) establish normal hyperbolicity of $\mathcal{M}_0$ for $\mathcal{V}_0$ in the $W_{m,2}$ geometry (the transverse spectrum is bounded away from the imaginary axis by the clustering of D^s ; this is where the Jacobian analysis of Tomihari and Karakida [2025] enters); (b) apply Fenichel-type persistence so that $\mathcal{M}_0$ perturbs to an invariant manifold $\mathcal{M}_\varepsilon$ on which the reduced dynamics are the drift of Proposition 12, giving a relative periodic orbit; (c) identify the rotation-invariant member μ_{ring} as a stationary measure of the full continuity equation with $\mathcal{V}_\varepsilon[\mu_{\text{ring}}]$ divergence-free, tangent to its support, and nonzero—a non-equilibrium steady state. Establishing (a)–(c) rigorously—where the delicate point is the infinite-dimensional persistence at the level of measures—is the principal analytical problem left open by this work.

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